MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

NETWORK, RESISTOR, PROGRAMMABLE, ANALOG, KEY CODE GGG

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic programmable resistor network module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.05 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be GGG.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/180-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

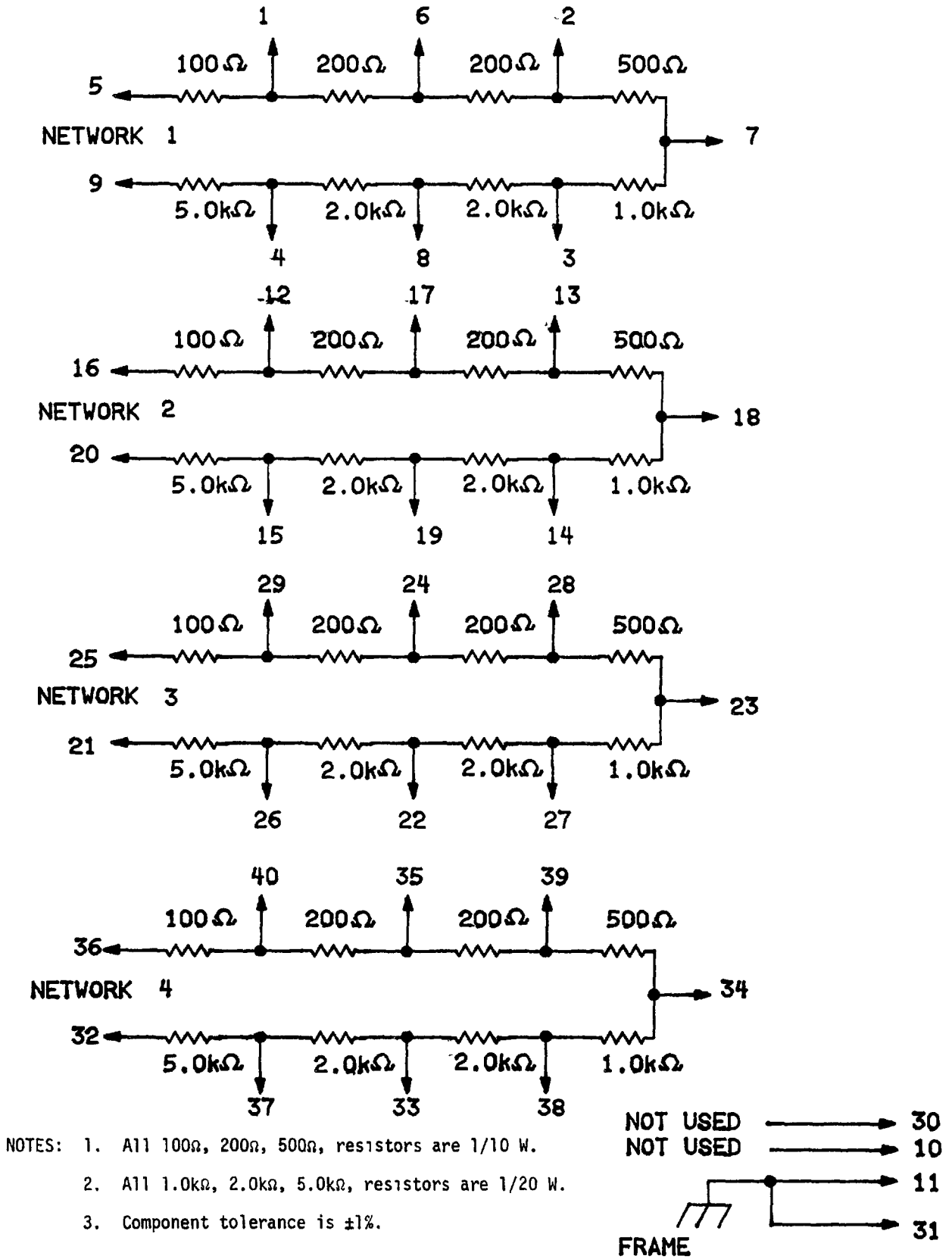
4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through E shall be performed in any order with the following exceptions. If test condition A is part of a sequence of tests, the order of testing shall be as follows: A where applicable, followed by the remaining tests in the sequence.

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NOTES: 1. All 100Ω, 200Ω, 500Ω, resistors are 1/10 W.

2. All 1.0kΩ, 2.0kΩ, 5.0kΩ, resistors are 1/20 W.

3. Component tolerance is ±1%.

FIGURE 1. GGG module function diagram, programmable resistor network.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A7	Isolation	II	X	100	∞	10	∞	100	∞	10	∞	MΩ
B1	Continuity, frame	III	S	0	1	0	1	0	1	0	1	Ω
B2	Continuity, pin to pin	III	X	0	1	0	1	0	1	0	1	Ω
C1-C4	Resistance	IV	X	99.3	100.8	99	101	99	101	99	101	Ω
C5-C12	Resistance	IV	X	198.5	201.5	198	202	198	202	198	202	Ω
C13-C16	Resistance	IV	X	496.2	503.8	495	505	495	505	495	505	Ω
C17-C20	Resistance	IV	X	993	1008	990	1010	990	1010	990	1010	Ω
C21-C28	Resistance	IV	X	1985	2015	1980	2020	1980	2020	1980	2020	Ω
C29-C32	Resistance	IV	X	4962	5038	4950	5050	4950	5050	4950	5050	Ω
C33-C36	Resistance	IV	Q	297.7	302.3	297	303	297	303	297	303	Ω
C37-C40	Resistance	IV	Q	496.2	503.8	495	505	495	505	495	505	Ω
C41-C44	Resistance	IV	Q	993	1008	990	1010	990	1010	990	1010	Ω

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
C45-C48	Resistance	IV	Q	1985	2015	1980	2020	1980	2020	1980	2020	Ω
C49-C52	Resistance	IV	Q	3970	4030	3960	4040	3960	4040	3960	4040	Ω
C53-C56	Resistance	IV	Q	5955	6045	5940	6060	5940	6060	5940	6060	Ω
C57-C60	Resistance	IV	Q	10.917	11.083	10.89	11.11	10.89	11.11	10.89	11.11	kΩ
D1-D4	Capacitance	V		0	30	0	30	0	30	0	30	pF
E1,E3-E5, E8-E10, E-12	Crosstalk	VI	S	0	1	0	1	0	1	0	1	mVrms
E2,E6,E7, E11	Crosstalk	VI	S	0.0	5.6	0.0	5.6	0.0	5.6	0.0	5.6	mVrms

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

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4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through E in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B through E shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 2. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests B2 and C through E shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through E shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/180-1 supersedes part number M28787/180. The two parts are interchangeable.

6.4 Module function. The module consists of four independent resistor networks as shown on figure 1. Resistor values, tolerances, and wattages are as specified on figure 1.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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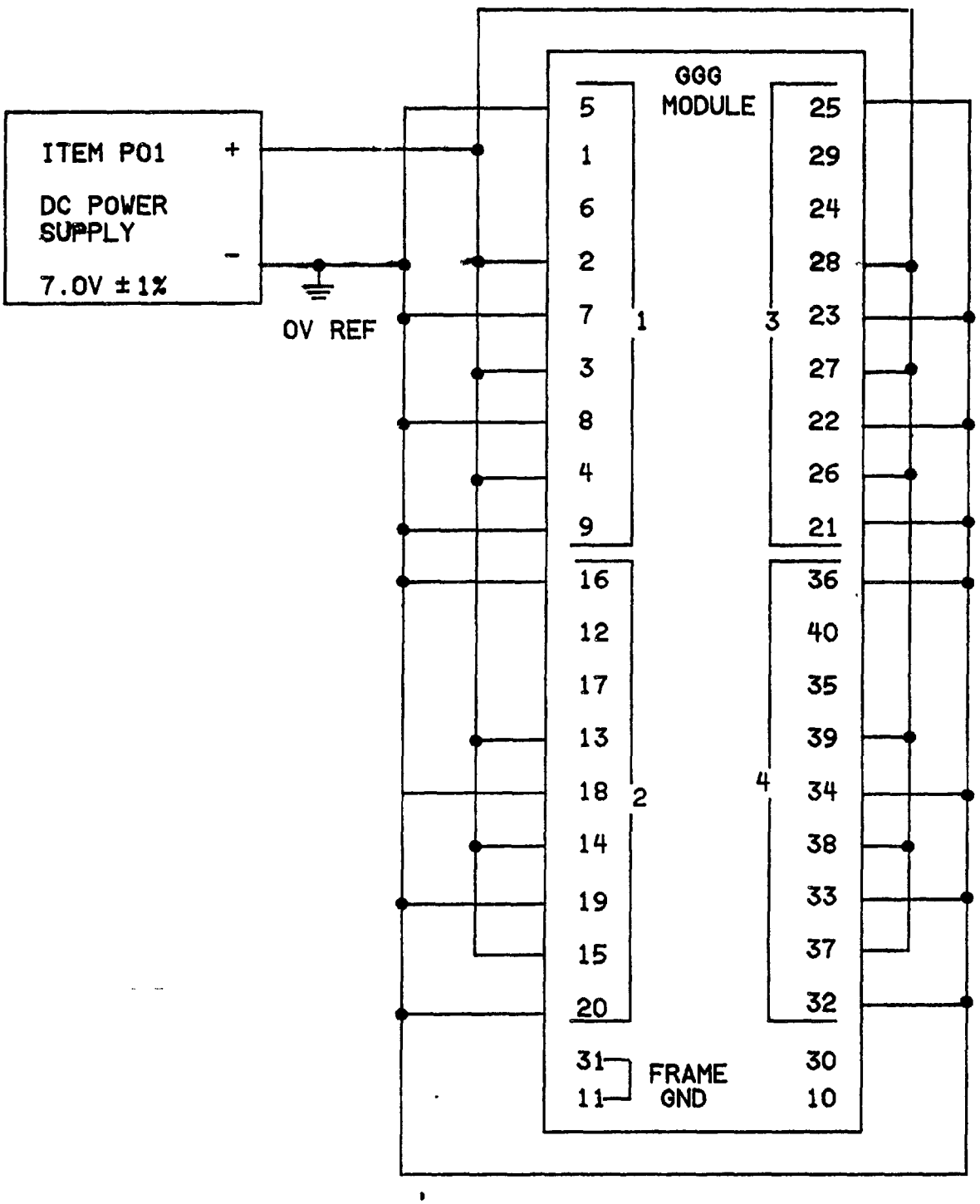


FIGURE 2. Operating temperature, component temperature, vibration, and life test circuit.

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TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	C,D,E,F,10,30	11,31	100	∞	MΩ	3
A2	↓	C,D,E,F,11,31	10,30	↓	↓	↓	↓
A3		10	30				
A4		C	D,E,F				
A5		D	C,E,F				
A6		E	C,D,F				
A7		F	C,D,E				

TABLE III. Continuity test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Continuity, frame	11	Frame	0	1	Ω	4
B2	Continuity, pin to pin	11	31	0	1	Ω	4

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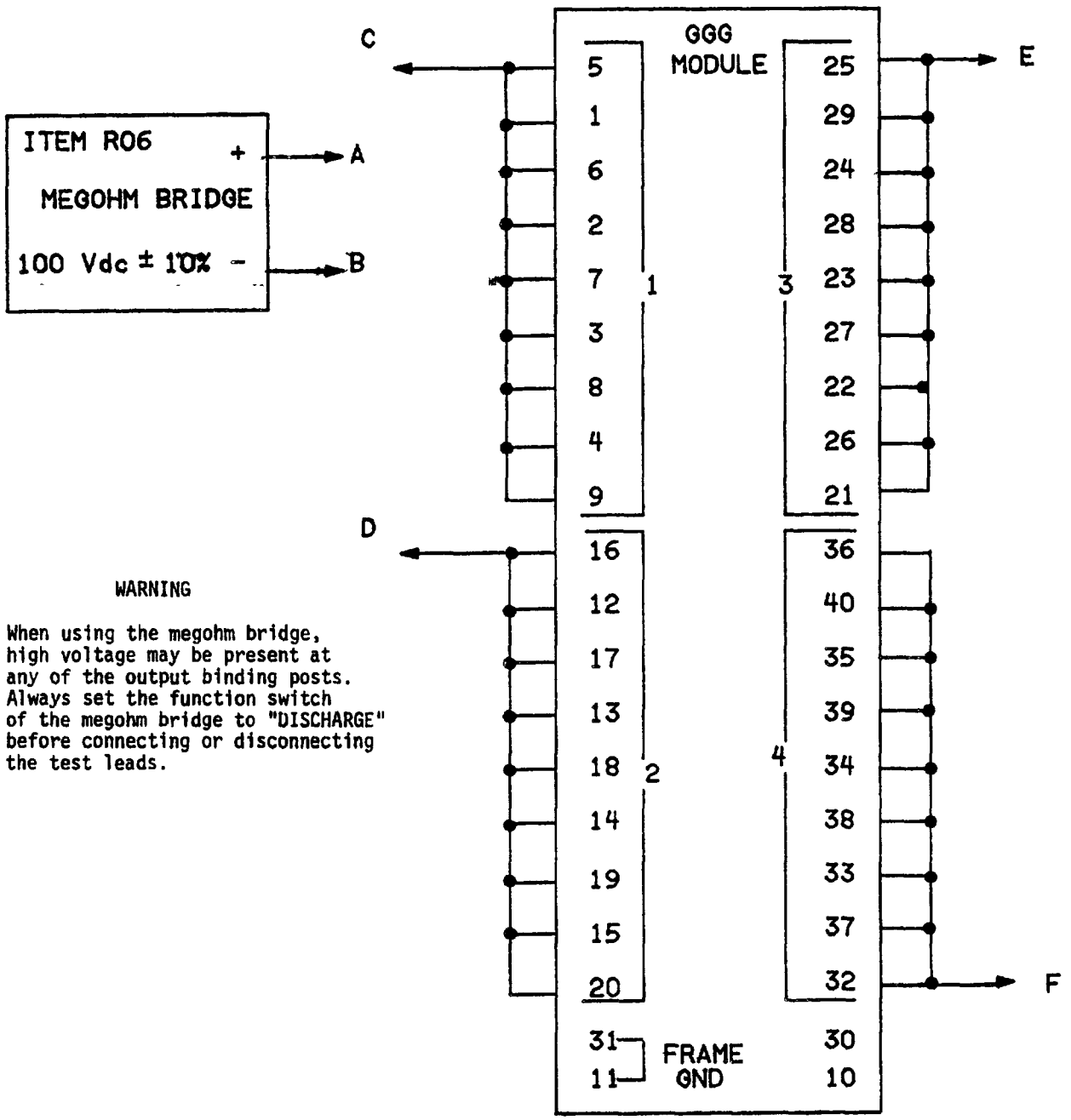
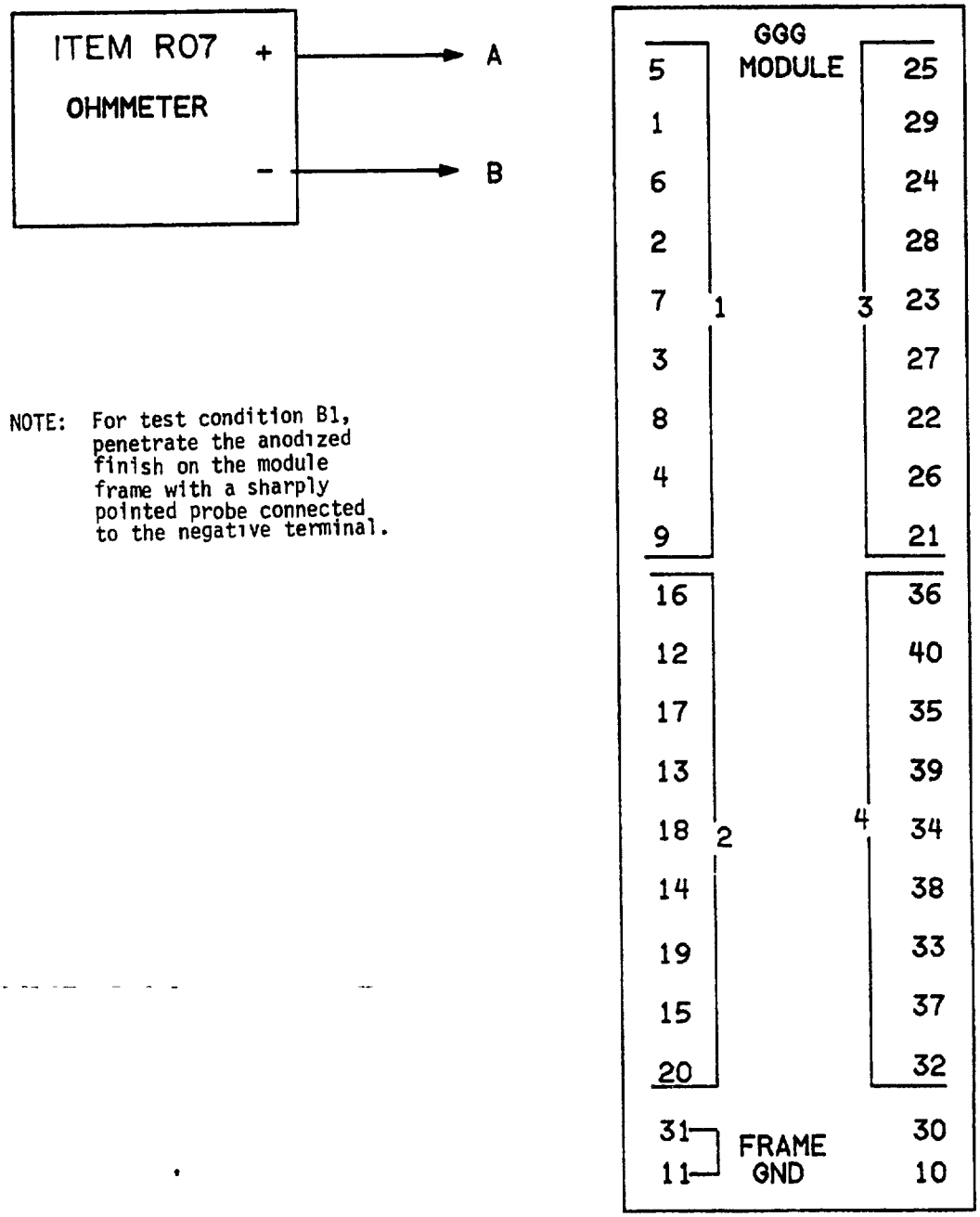


FIGURE 3. Isolation test circuit for table II.

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NOTE: For test condition B1, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 4. Continuity test circuit for table III.

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TABLE IV. Resistance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Resistance	5	1	99.3	100.8	Ω	5
C2		16	12				
C3		25	29				
C4		36	40				
C5		1	6	198.5	201.5		
C6		6	2				
C7		12	17				
C8		17	13				
C9		29	24				
C10		24	28				
C11		40	35				
C12		35	39				
C13		2	7	496.2	503.8		
C14		13	18				
C15		28	23				
C16		39	34				
C17		7	3	993	1008		
C18		18	14	993	1008		
C19		23	27	993	1008		

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TABLE IV. Resistance test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C20	Resistance	34	38	993	1008	Ω	5
C21		3	8	1985	2015		
C22		8	4				
C23		14	19				
C24		19	15				
C25		27	22				
C26		22	26				
C27		38	33				
C28		33	37				
C29		4	9	4962	5038		
C30		15	20				
C31		26	21				
C32		37	32				
C33		5	6	297.7	302.3		
C34		16	17				
C35		25	24				
C36		36	35				
C37		5	2	496.2	503.8		
C38		16	13	496.2	503.8		
C39		25	28	496.2	503.8		

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TABLE IV. Resistance test - continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C40	Resistance	36	39	496.2	503.8	Ω	5
C41	↓	5	7	993	1008	↓	↓
C42		16	18	↓	↓		
C43		25	23				
C44		36	34				
C45		5	3	1985	2015		
C46		16	14	↓	↓		
C47		25	27				
C48		36	38				
C49		5	8	3970	4030		
C50		16	19	↓	↓		
C51		25	22				
C52		36	33				
C53		5	4	5955	6045		
C54		16	15	↓	↓		
C55		25	26				
C56		36	37				
C57		5	9	10.917	11.083	kΩ	
C58		16	20	↓	↓	↓	
C59		25	21				
C60	↓	36	32				↓

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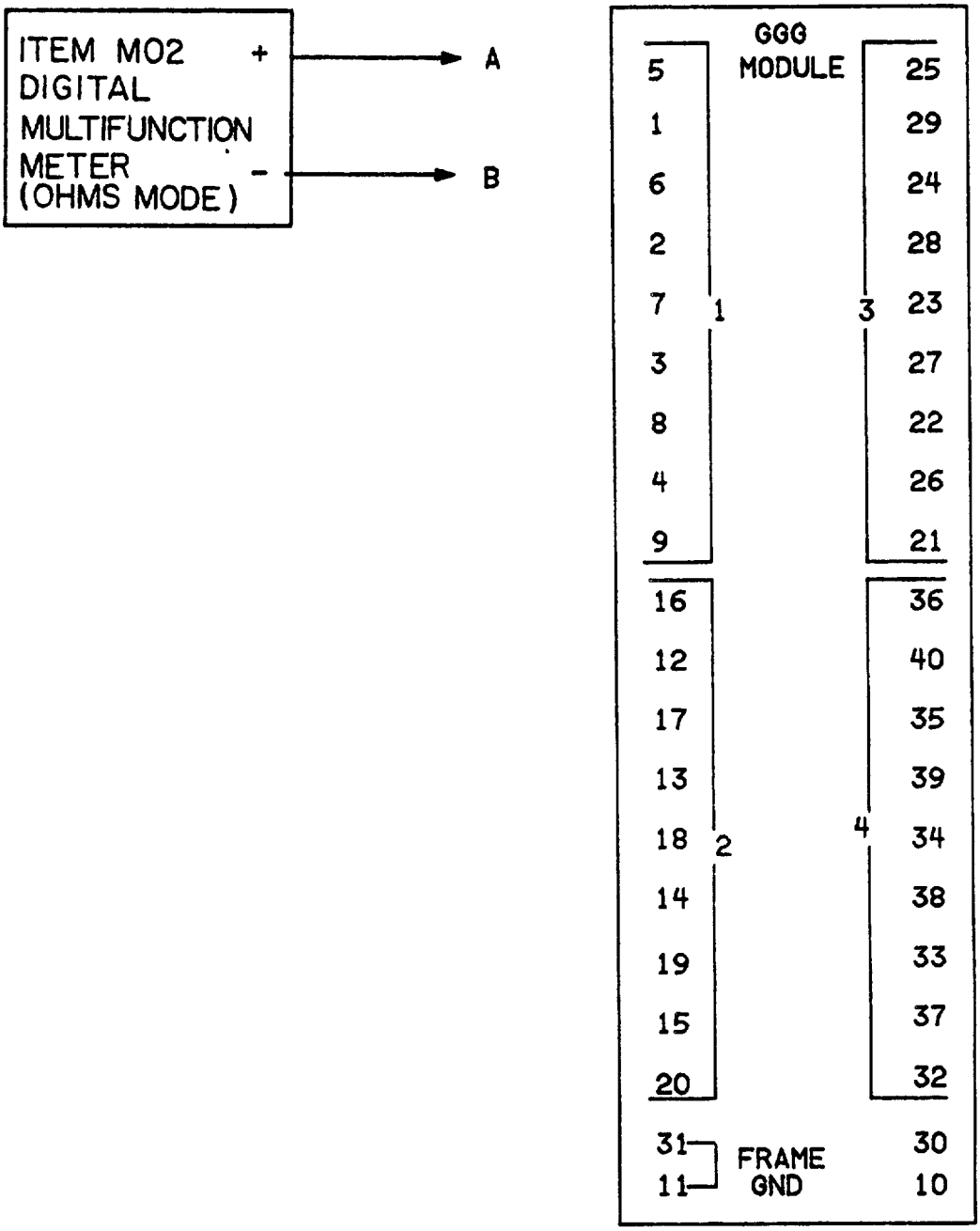


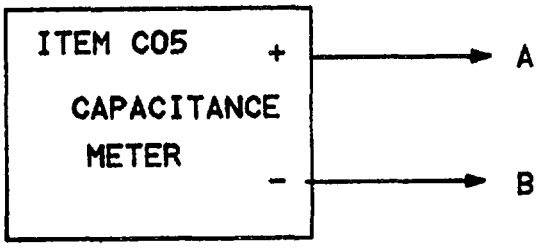
FIGURE 5. Resistance test circuit for table IV.

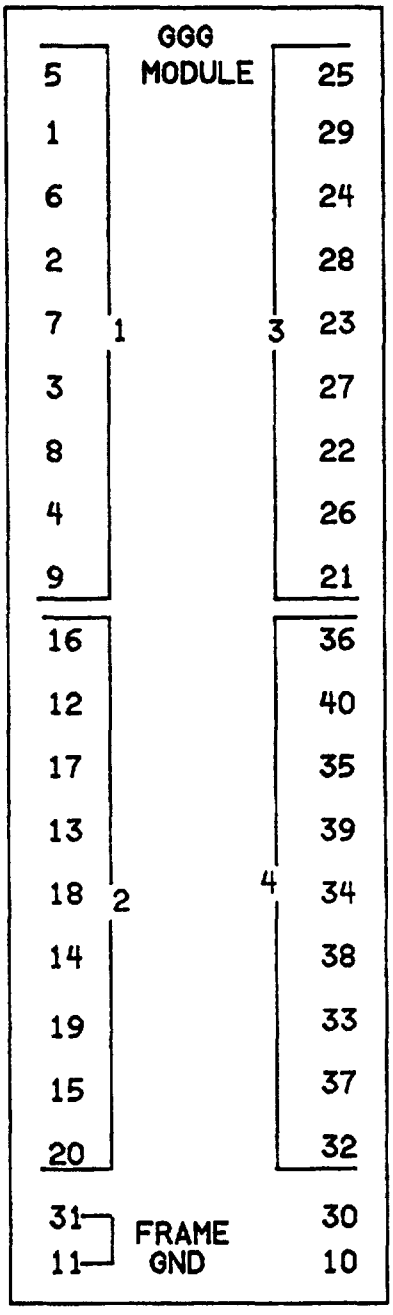
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TABLE V. Capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	Capacitance	7	18,23,34	0	30	pF	6
D2	↓	18	7,23,34	↓	↓	↓	↓
D3		23	7,18,34				
D4		34	7,18,23				

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NOTES:
1. Null out the effect of the test leads prior to connecting to the module.

2. Do not attach the test connector or wires to any of the module pins other than pins under test. Stray capacitance added to other pins may affect the readings.

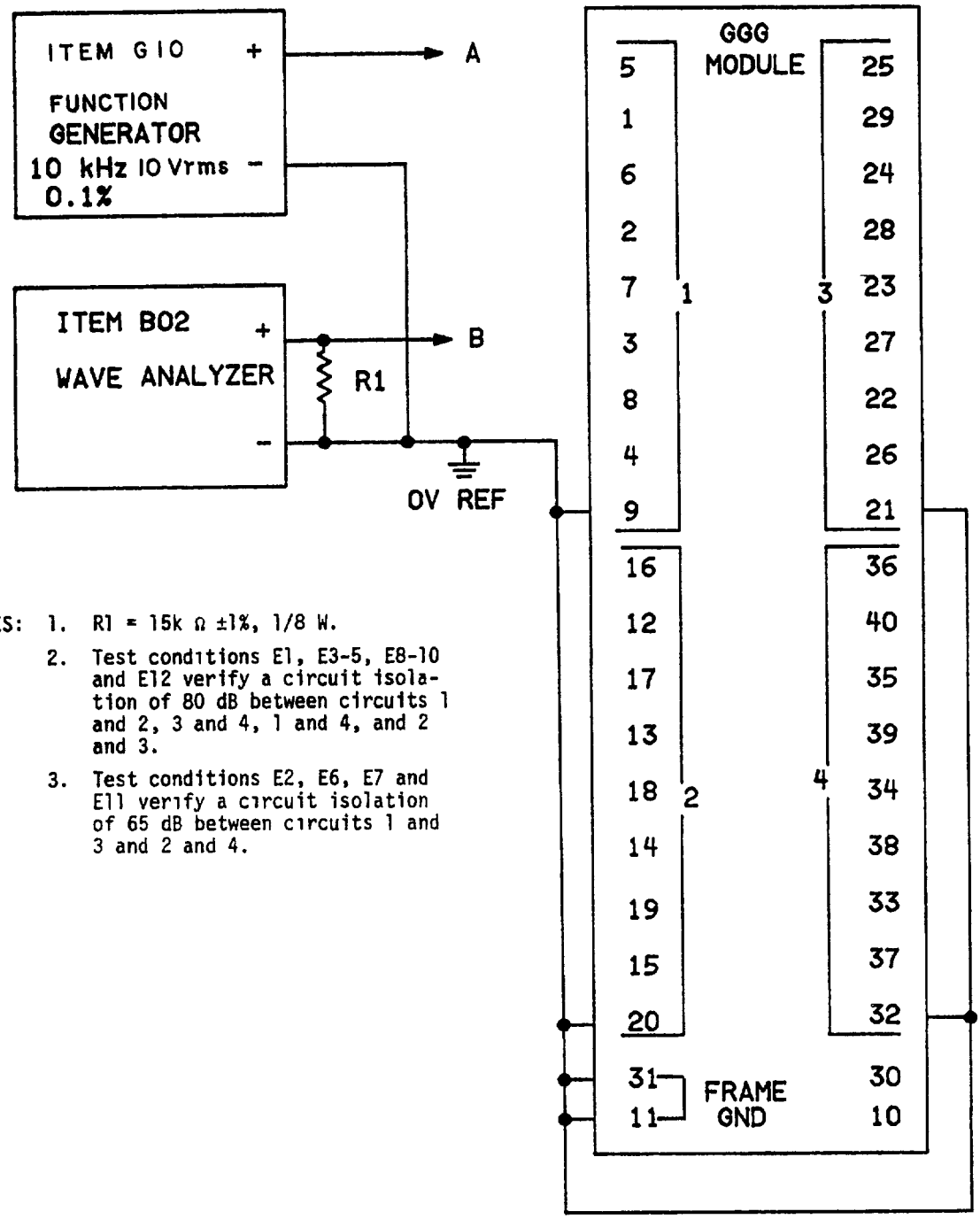
FIGURE 6. Capacitance test circuit for table V.

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TABLE VI. Crosstalk test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	Crosstalk	16	5	0	1.0	mVrms	7
E2		25	5		5.6		
E3		36	5		1.0		
E4		5	16		1.0		
E5		25	16		1.0		
E6		36	16		5.6		
E7		5	25		5.6		
E8		16	25		1.0		
E9		36	25		1.0		
E10		5	36		1.0		
E11		16	36		5.6		
E12		25	36		1.0		

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NOTES:
1. R1 = 15k Ω ±1%, 1/8 W.
2. Test conditions E1, E3-5, E8-10 and E12 verify a circuit isolation of 80 dB between circuits 1 and 2, 3 and 4, 1 and 4, and 2 and 3.
3. Test conditions E2, E6, E7 and E11 verify a circuit isolation of 65 dB between circuits 1 and 3 and 2 and 4.

FIGURE 7. Network crosstalk test circuit for table VI.

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